

Harbor Ridge V1

Source-System Map

Healthcare Executive Operating System (HEOS)

Document Field	Value
Version	1.0
Status	FINAL — Approved for Harbor Ridge V1
Project	Healthcare Executive Operating System (HEOS)
Artifact	Harbor Ridge V1
Purpose	Define the minimum source systems, handoffs, identifiers, identity-resolution logic, evidence-trust principles, and representative degradation points required to trace patient acquisition through inquiry, admission, clinical care, billing, and collected revenue.

1. Purpose

Harbor Ridge V1 models the patient acquisition and revenue pathway of a fictional behavioral healthcare organization. The Source-System Map defines where information originates, how it moves between systems, which identifiers preserve lineage, how multiple human contacts resolve to a single patient opportunity, and where that lineage can degrade.

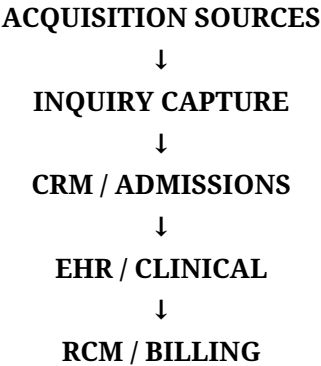
The architecture is designed around one central executive question:

Can Harbor Ridge trace a patient outcome and collected revenue backward through the organization to the acquisition evidence that originally generated the opportunity?

V1 is not intended to reproduce a production healthcare technology stack. It establishes the minimum architecture necessary to analyze the acquisition-to-revenue pathway reliably.

2. Executive Architecture

The Harbor Ridge acquisition-to-revenue pathway consists of six primary system layers:



**COLLECTIONS / FINANCIAL OUTCOME**

The analytical layer sits across these systems:

SOURCE SYSTEMS**NORMALIZED HARBOR RIDGE DATA****AI / ANALYTICAL REASONING****EXECUTIVE DECISION SUPPORT**

The objective is to preserve an auditable identity, attribution, and outcome chain across every major handoff.

3. Source-System Inventory

Layer	Source System / Channel	Primary Purpose	Representative Data
Acquisition	Google Ads	Paid search acquisition	Campaign, ad group, keyword, search term, spend, click identifiers
Acquisition	Microsoft Ads	Paid search acquisition	Campaign, ad group, keyword, search term, spend, click identifiers
Acquisition	Meta Ads	Paid social acquisition	Campaign, ad set, creative, audience, spend, delivery metrics
Acquisition	SEO / Organic Search	Organic acquisition	Query, landing page, clicks, impressions, geographic context
Acquisition	Google Business Profile / Local Search	Local discovery and calls	Calls, website visits, local search interactions
Acquisition	Professional Referral / Business Development	Professional referral acquisition	Professional account, outreach representative, referral event
Inquiry Capture	Website / Web Forms	Capture consumer inquiries	Form start, completion, UTM parameters, click identifiers
Inquiry Capture	Call Tracking	Capture inbound phone inquiries	Tracking number, source, timestamp, duration
Admissions	CRM	Manage inquiries, identity resolution, opportunities, VOB	Inquiry, contact role, opportunity, source, VOB, referral relationship
Clinical	EHR	Manage patient episode of care	Admission, level of care, discharge, clinical episode
Revenue	RCM / Billing	Manage claims and reimbursement	Claim, payer, DOS, allowed amount, payment, denial
Financial	Collections / Remittance	Verify realized financial outcome	Insurance payment, patient payment, adjustments, collected cash

4. Acquisition Architecture

4.1 Digital Acquisition

Digital acquisition may originate from Google Ads, Microsoft Ads, Meta Ads, Organic Search, Google Business Profile / Local Search, or direct website activity. Where available, acquisition metadata should be captured before or at the point an inquiry enters the CRM.

- GCLID or equivalent advertising click identifier
- UTM source / medium / campaign
- Campaign ID
- Ad Group / Ad Set ID
- Keyword
- Search Term
- Landing Page
- Call Tracking Source

Advertising platforms are not treated as systems of record for business outcomes. Platform conversions describe platform-observed activity; CRM inquiries, VOBs, admissions, and downstream financial outcomes describe business activity.

4.2 Professional Referral / Business Development

Professional referrals require a separate attribution model because the originating relationship may not correspond to the patient's final digital touchpoint.

- Professional Account
- Referring Professional
- Outreach Representative
- Referral Event
- Patient Opportunity

A patient referred by a psychiatrist may later search for Harbor Ridge and submit a web form. The web session is the arrival mechanism; the professional relationship may be the originating acquisition source. Harbor Ridge V1 preserves both when the evidence supports them.

5. CRM / Admissions Layer

The CRM is the primary operational system for inquiries before clinical admission. Harbor Ridge distinguishes an individual inquiry from the broader Patient Opportunity to which one or more inquiries may belong.

Every Patient Opportunity receives a canonical identifier:

PATIENT_OPPORTUNITY_ID / CRM_OPPORTUNITY_ID

↓

Canonical format: HRO-#####



Example: HRO-000184

Representative CRM fields include:

- CRM Opportunity ID
- Inquiry Timestamp
- Acquisition Source and Channel
- Campaign ID
- Professional Referral Account ID
- Outreach Representative ID
- Contact Role
- Identity Match Confidence
- VOB Submitted
- VOB Outcome
- Payer
- Admission Financial Status
- Admission Status

The CRM is the preferred system of record for determining whether a platform event resulted in an actual human inquiry.

5.1 Identity Resolution and Patient Opportunity

Behavioral-health inquiries are not reliably one person, one call, one opportunity. A loved one may call first, another family member may follow, and the patient may later contact Harbor Ridge directly. These are distinct inquiry events that can belong to one Patient Opportunity.

INQUIRY: Mom calls Monday



INQUIRY: Dad calls Tuesday



INQUIRY: Patient calls Wednesday



PATIENT OPPORTUNITY: HRO-000184

Contact roles include:

- Patient
- Loved One
- Professional Referral Source

Identity-resolution confidence is represented as:

Match State	Meaning
Confirmed	Evidence is sufficient to treat the records as the same Patient Opportunity.
Probable	Strong evidence supports a match, but a deterministic identifier is absent.
Possible	Some evidence supports a relationship, but human review is required.
Unmatched	No defensible linkage has been established.

Identity resolution links records without erasing their original history. Each inquiry retains its timestamp, arrival mechanism, acquisition evidence, and contact role even when multiple inquiries resolve to one Patient Opportunity.

6. Verification of Benefits and Admission Financial Status

Verification of Benefits (VOB) is the principal financial-viability checkpoint between inquiry and admission. It asks:

Did the inquiry possess insurance coverage compatible with Harbor Ridge's payer strategy?

Representative VOB outcomes:

- Pending
- Viable
- Non-Viable
- Unable to Verify

VOB outcome is analytically separate from the organization's admission decision. Harbor Ridge therefore preserves an additional Admission Financial Status:

- Financially Cleared
- At-Risk Admission
- Not Financially Cleared

An At-Risk Admission represents a clinically appropriate admission made while reimbursement remains unresolved or uncertain. Financial viability informs the admission pathway but does not mechanically dictate every admission decision.

7. CRM → EHR Handoff

Once a patient is admitted, the system of record transitions from the CRM to the EHR, which creates an episode identifier such as:

EHR_EPISODE_ID



Example: KIPU-9921

The critical relationship is:

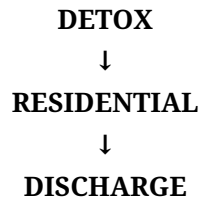
CRM_OPPORTUNITY_ID ↔ EHR_EPISODE_ID

Whenever technically possible, the CRM Opportunity ID should persist into the EHR. Without it, Harbor Ridge may know a patient admitted without reliably knowing which Patient Opportunity and acquisition evidence generated the admission.

8. EHR / Clinical Layer

The EHR becomes the system of record for the clinical episode.

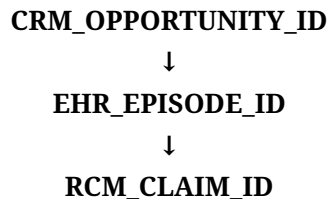
- EHR Episode ID
- Admission Date / Time
- Program
- Level of Care
- Treatment Location
- Payer
- Authorization Window
- Level-of-Care Transfer
- Discharge Date
- Discharge Disposition



Harbor Ridge V1 must distinguish a true new admission from a level-of-care transition or administrative discharge/re-admit event. Otherwise, one patient episode may incorrectly appear as multiple independent admissions.

9. EHR → RCM Handoff

The RCM system introduces another system boundary and creates identifiers such as RCM_CLAIM_ID.

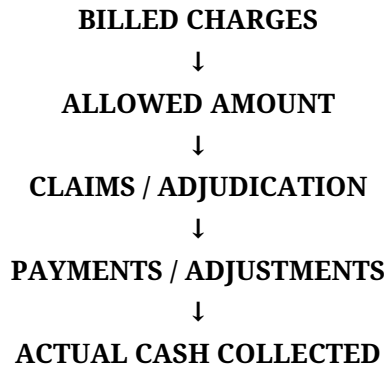


- Claim ID
- Patient / Episode Identifier
- Dates of Service
- Payer
- Billed Charges
- Allowed Amount
- Denial Status
- Adjustments
- Insurance Payments
- Patient Responsibility
- Payment Date

A single clinical episode may produce multiple claims. One admission therefore does not necessarily equal one claim; claims remain associated with the originating episode.

10. Collections / Financial Outcome

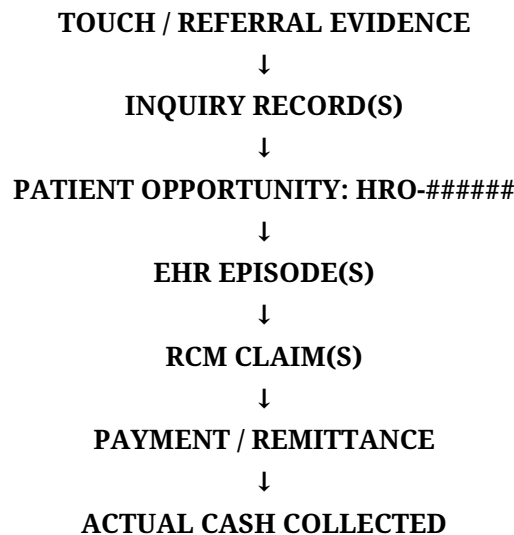
Harbor Ridge distinguishes accounting estimates from realized financial outcomes.



For retrospective acquisition analysis, ACTUAL_CASH_COLLECTED is preferred over billed charges or expected reimbursement. Recent cohorts should not be compared directly with mature cohorts without accounting for collection lag.

11. The Harbor Ridge Golden Thread

The Golden Thread is the identity, attribution, and outcome lineage that connects messy upstream human behavior to structured downstream healthcare operations.



The relationships are not assumed to be one-to-one. A Patient Opportunity may contain multiple touches and inquiry records; an EHR episode may generate multiple claims; claims may generate multiple payments and adjustments.

Which acquisition sources generated financially viable opportunities, admissions, clinical episodes, and ultimately collected revenue?

Harbor Ridge V1 identifies where this thread is intact, missing, or dependent on weaker matching logic.

12. Major V1 Failure Points and Degradation Types

Harbor Ridge uses four canonical degradation classes: Observability Loss, Identity Loss, Attribution Loss, and Outcome-Linkage Loss. The six failures below are representative V1 examples of those classes, not an exhaustive production failure catalog.

#	Handoff / Area	Representative Failure	Degradation Type	Executive Consequence
1	Acquisition → Inquiry	A tracking event or metadata is never captured before CRM creation	Observability Loss	Demand or source evidence cannot be observed reliably
2	Inquiry / CRM Identity	Mom, dad, and patient remain separate unrelated records for one underlying opportunity	Identity Loss	One patient opportunity is overstated, fragmented, or double-counted
3	Professional Referral → CRM	Referring professional is not linked to the Patient Opportunity	Attribution Loss	BD contribution is credited to digital, direct, or unknown
4	Digital Acquisition → CRM	Acquisition metadata is lost or overwritten	Attribution Loss	Lead exists but source becomes Unknown / Direct
5	CRM → EHR / EHR → RCM	Opportunity or episode identity is lost or fragmented	Outcome-Linkage Loss	Admission or revenue cannot reliably map to the originating opportunity
6	RCM → Collections	Payments cannot be associated with the originating claim / episode	Outcome-Linkage Loss	Collected cash becomes unattributed to acquisition source

Additional production-level failure scenarios are outside V1 unless required for the synthetic dataset or analysis to function correctly.

13. System-of-Record Hierarchy

Business Question	Preferred System of Record
What advertising activity occurred?	Advertising Platform
What organic search activity occurred?	Google Search Console / Analytics
Did a real inquiry enter the business?	CRM
Which inquiries belong to the same Patient Opportunity?	CRM identity-resolution record
Was a VOB performed?	CRM / Admissions Workflow
Who referred the opportunity?	CRM Professional Referral Record
Did the patient admit?	EHR
What level of care did the patient receive?	EHR
What was billed?	RCM
What was paid?	RCM / Remittance
What cash was actually collected?	Financial / Collections Record

The system closest to the actual operational event is preferred over a downstream estimate or upstream proxy.

13.1 Evidence Provenance and Trust

System ownership and evidence confidence are separate dimensions. A field can reside in the correct system of record and still differ in reliability depending on how it was produced.

Evidence Class	V1 Interpretation
System-observed	Mechanically recorded event or system timestamp; generally strongest for proving that an event occurred.
Human-entered	Operationally valid but subject to workflow, interpretation, omission, and data-entry error.
AI-derived / Inferred	Analytical interpretation or match suggestion; useful for reasoning but never treated as source truth.

Harbor Ridge also preserves the distinction between hard-gate dispositions and softer behavioral or judgment-based classifications. Detailed evidence-provenance and trust-tier rules remain documented in docs/source-system-architecture-notes.md; this map carries forward the governing principle without duplicating that full framework.

14. Analytical Layer

Harbor Ridge V1 will normalize selected source-system data into an analytical dataset. The analytical layer is not itself a source system.

- Which acquisition channels generate viable VOBs?
- Which campaigns generate admissions rather than platform conversions?
- Which professional accounts generate legitimate opportunities?
- Where are duplicate inquiries fragmenting a Patient Opportunity?
- Where are inquiries being lost between marketing and admissions?
- Where are admissions being lost between VOB and physical intake?
- Which clinical episodes generate mature collected revenue?
- Where has the Golden Thread degraded, and by which degradation class?
- Which operational bottleneck best explains a change in census or financial performance?

The AI layer reasons over normalized evidence rather than replacing the underlying source systems or promoting inferred evidence to source truth.

15. V1 Scope Boundary

Harbor Ridge V1 is an executive decision-support portfolio project. It is not intended to reproduce:

- A production EHR
- A production CRM
- A medical billing clearinghouse
- A complete healthcare data warehouse
- A production patient identity-resolution engine

- A production HIPAA infrastructure
- A full clinical decision-support system

V1 contains only enough architecture and synthetic data to demonstrate:

1. Source-system lineage
2. Acquisition attribution
3. Patient-opportunity identity resolution
4. Funnel reconstruction
5. Data-quality degradation detection
6. Operational diagnosis
7. Revenue attribution
8. AI-assisted executive reasoning

Anything beyond these requirements is deferred unless necessary for the V1 demonstration.

16. Definition of Success

The Source-System Map is successful when a human reviewer or LLM can understand:

1. Where Harbor Ridge data originates.
2. How one or more inquiries resolve to a Patient Opportunity.
3. How acquisition evidence and arrival mechanism can coexist.
4. How an opportunity becomes an admission.
5. How an admission becomes a clinical episode.
6. How an episode becomes one or more claims.
7. How claims become collected cash.
8. Which identifiers connect those systems.
9. How Observability, Identity, Attribution, and Outcome-Linkage Loss differ.
10. Which system should be trusted for each major business question.
11. How evidence provenance affects confidence.
12. How the resulting evidence supports executive decision-making.

Once these relationships are sufficiently represented, the Source-System Map is considered complete for Harbor Ridge V1.